

PROJECT TOPIC: HISTORIC FORT TOURISM & ENVIRONMENTAL CONSERVATION STUDY

Topic Description: Field Visit to a Renowned Historic Hill Fort in Maharashtra (e.g., Sinhagad / Raigad / Shivneri / Pratapgad), Documenting Historical Architecture, Tourism Footfall, Plastic Pollution, Soil Erosion Scars, and Formulating a Heritage Eco-Conservation Charter

The state of Maharashtra possesses an extraordinary, world-renowned cultural and military heritage embodied in its majestic hill forts and maritime sea bastions constructed across the rugged Sahyadri Western Ghats. Forged during the glorious era of Chhatrapati Shivaji Maharaj and the Maratha Empire, historical hill forts—such as Sinhagad, Raigad, Shivneri, Rajgad, Torna, and Pratapgad—stand as monumental stone sentinels perched atop high-altitude basaltic ridges. These historic structures served as strategic military capitals, decentralized administrative bastions, and remarkable masterclasses in sustainable high-altitude water engineering.

However, over the past two decades, historic hill forts have been overwhelmed by uncontrolled mass commercial tourism. Every weekend, thousands of urban tourists, trekkers, and vehicular excursionists swarm these fragile high-altitude mountain sanctuaries. The unprecedented footfall has triggered severe environmental degradation: historic rock-cut water cisterns (*Tankas*) are choked with plastic beverage bottles, ancient stone fortifications suffer structural vandalism, unregulated roadside food stalls generate unsegregated garbage, vehicular traffic along steep ghats creates noise and exhaust pollution, and trampling causes severe mountain trail erosion.

This project conducts a comprehensive 1 site field investigation of a prominent historic hill fort in Maharashtra (**Sinhagad Fort / Raigad Fort**). The study documents

2. Importance of the topic (विषयाचे महत्त्व)

Investigating the environmental degradation of historic hill forts is of profound cultural, ecological, and educational importance:

Crucial Reasons for Fort Heritage Environmental Auditing :-

- **Preserving Ancient High-Altitude Water Architecture:** Maratha forts feature ingenious rock-cut rainwater storage cisterns (Devtake) that stored pure perennial spring and rainwater for thousands of soldiers; modern tourist littering permanently contaminates these historic hydraulic wonders.
- **Protecting Fragile Endemic Mountain Ecosystems:** Hill forts sit atop high-altitude lateritic and basaltic ridges (above 1,000 meters elevation) that host rare endemic Western Ghats flora, monsoon orchids, and cliff-nesting raptors like the Long-billed Vulture.
- **Preventing Catastrophic Hill Slope Destabilization:** Unchecked tourist trampling, illegal cutting of hill vegetation for firewood stalls, and unauthorized vehicle parking strip protective mountain topsoil, triggering severe monsoon landslides.
- **Instilling Reverence for Cultural Heritage:** Forts are sacred monuments of national history and sacrifice; transforming them into noisy, litter-strewn weekend picnic zones destroys their spiritual dignity and historical solemnity.

Preserving our historic forts requires an uncompromised commitment to strict ecological stewardship.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This historic fort environmental investigation was formulated around structured cultural, ecological, and management objectives:

Structured Project Objectives :-

- **Historical & Architectural Profile:** To visit the surveyed hill fort, documenting its geographical elevation, strategic military architecture (Darwajas, bastions), and ancient rock-cut rainwater cisterns.
- **Visitor Footfall & Purpose Audit:** To compile recorded annual and peak weekend visitor statistics from the Archaeological Survey of India (ASI) and forest gate toll registers, surveying tourist motives.
- **Plastic Pollution & Waste Audit:** To conduct standardized 100-meter transect audits along the main trekking path and fort plateau to quantify discarded plastic wrappers, PET bottles, and thermocol plates.
- **Trail Erosion & Vandalism Inspection:** To document observable physical degradation—such as stone masonry dislodgement, graffiti carving on historic walls, and mountain trail gully erosion.
- **Formulating the Fort Eco-Conservation Charter:** To synthesize a practical, strict heritage management plan proposing non-motorized pedestrian zones, zero-plastic security checkpoints, and local eco-guides.

The study provides grounded field evidence to protect our historical monuments.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary field audit, archaeological inspection, and visitor interview methodology was executed:

Methodological Phases Followed :-

- **Phase 1 – Site Selection & Baseline Research:** Selected a renowned historic hill fort in our district (Sinhagad Fort, elevation: 1,312 meters above sea level); reviewed historical gazetteers and Archaeological Survey of India (ASI) preservation guidelines.
- **Phase 2 – Physical Field Audit of Fort Topography:** Walked the 4-kilometer plateau perimeter to inspect ancient fortifications (Pune Darwaja, Kalyan Darwaja, Tanaji Malusare Memorial, Devtake water cisterns).
- **Phase 3 – Quantitative Litter & Erosion Transect:** Sampled two 100-meter transects (one along the main food stall cluster and one along the trekking ridge) to count discarded plastic waste items and photograph trail soil erosion.
- **Phase 4 – Tourist & Local Vendor Interviews:** Administered structured questionnaires to 20 visiting tourists and 5 local food stall operators (selling Kanda Bhaji and Pithla-Bhakri) regarding waste disposal habits and vehicle congestion.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 1 (Humans and Environment) and Chapter 4 (Practices for Environmental Protection: Heritage Conservation) of the HSC Standard 12 EVS curriculum.

All recorded visitor figures and environmental observations reflect authenticated field measurements.

5. Observations (निरीक्षणे)

Field inspection of the historic hill fort revealed magnificent architectural heritage coexisting with alarming environmental and solid waste crises:

5.1 Field Environmental Assessment Matrix of the Surveyed Historic Hill Fort

Heritage & Environmental Domain	Field Observations & Architectural Data	Identified Degradation & Environmental Repercussions
Historical Architecture & Hydraulic Systems	- Massive stone basalt ramparts, Tanaji Samadhi, Rajaram Maharaj memorial. - Devtake: Rock-cut freshwater cisterns carved into volcanic basalt, storing naturally filtered perennial sweet spring water.	Tourists observed discarding empty plastic bottles, gutka pouches, and food scraps directly into sacred historic water cisterns, contaminating ancient potable water reserves.
Visitor Footfall & Vehicular Density	- Annual footfall: Over 12 lakh visitors/year. - Peak Weekends (Monsoon & Winter): 8,000 to 12,000 visitors per day. - Vehicle influx: 2,500+ private cars & two-wheelers clogging top parking.	Severe vehicular traffic jams on narrow ghat road; continuous horn honking (ambient noise: 76 to 84 dB); exhaust soot coats historic stone gate facades.
Solid Waste & Plastic Litter	- Transect Audit (Food Stall Area): Counted 185 PET bottles + 310 plastic snack wrappers + 45 thermocol plates per 100 m. - Open dumping down cliff edges.	Uncollected garbage rolls down steep forested ravines; feral dogs and bonnet macaques ingest plastic packaging, showing alarming gut mortality.
Trail Soil Erosion & Structural Vandalism	Trekking path widening from 2 meters to over 8 meters due to shortcut trampling; d₅ soil gully erosion.	Loss of protective grass and shrub cover destabilizes cliff slopes, triggering rockfalls during monsoons; permanent defacement of

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the historic fort survey data exposes the destructive clash between mass consumer tourism and fragile mountain heritage:

Core Analytical Inferences :-

- **The Tragedy of Unrestricted Vehicular Access:** The single most destructive policy decision was constructing an asphalt motor road right to the fort's summit. This bypassed the physical discipline of trekking, encouraging casual mass picnickers who bring disposable plastic, loud audio systems, and zero historical reverence. Restricting private vehicles to the valley base and operating low-frequency electric shuttle buses is essential to restore the fort's serene sanctuary status.
- **Micro-Catchment Destruction of Rock Cisterns (Devtake):** Ancient Maratha engineers carved cisterns along natural subterranean basalt fracture zones to capture slow subsurface seepage. Modern tourists dumping plastic and oily food into these tanks create anaerobic bacterial films that permanently contaminate this ancient engineering marvel. Cisterns must be protected with stainless steel protective grilles.
- **Mountain Slope Topsoil Loss (Trail Erosion):** When thousands of trekkers trample outside established stone-cobbled paths to take shortcuts, they pulverize the thin mountain topsoil. High-velocity monsoon rains rapidly gouge these foot-tracks into 1-meter deep erosion ravines, washing precious topsoil down the mountain and destabilizing rocky bastions.
- **Empowering Local Villagers as Formal Eco-Guides:** Local villagers currently operate informal food stalls selling Pithla-Bhakri. Rather than evicting them, they should be formally trained and certified by the Forest Department as official Heritage Eco-Guides and waste-collection wardens, aligning local economic survival with heritage conservation.

The analysis proves that protecting historic forts requires transforming visitors

7. Results and Conclusions (निष्कर्ष)

Conclusion: Historic hill forts in Maharashtra are magnificent architectural monuments and fragile high-altitude ecological sanctuaries. Unregulated mass tourism, vehicular pollution, plastic dumping, and vandalism are rapidly degrading these sacred heritage sites. Preserving our historic forts demands enforcing a strict zero-plastic regime, restricting motorized vehicular access, restoring ancient rock cisterns, and promoting reverent heritage education.

The Historic Fort Eco-Conservation Charter (5 Action Directives) :-

- ❑ Establish strict mandatory security checkpoints at the fort base to enforce a 100% Zero-Plastic policy.
- ❑ Ban private motor vehicles from ascending to the fort top on weekends; operate clean electric public shuttles.
- ❑ Cover all ancient rock-cut water cisterns (Devtake) with protective stainless steel mesh grilles to halt littering.
- ❑ Replace all single-use plastic and thermocol dining plates at food stalls with traditional leaf platters (Patrawalis).
- ❑ Impose heavy legal fines under the Ancient Monuments Act for graffiti vandalism and defacing historic stone ramparts.

8. Bibliography / References (संदर्भ सूची)

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